

Enterprise Worker Safety



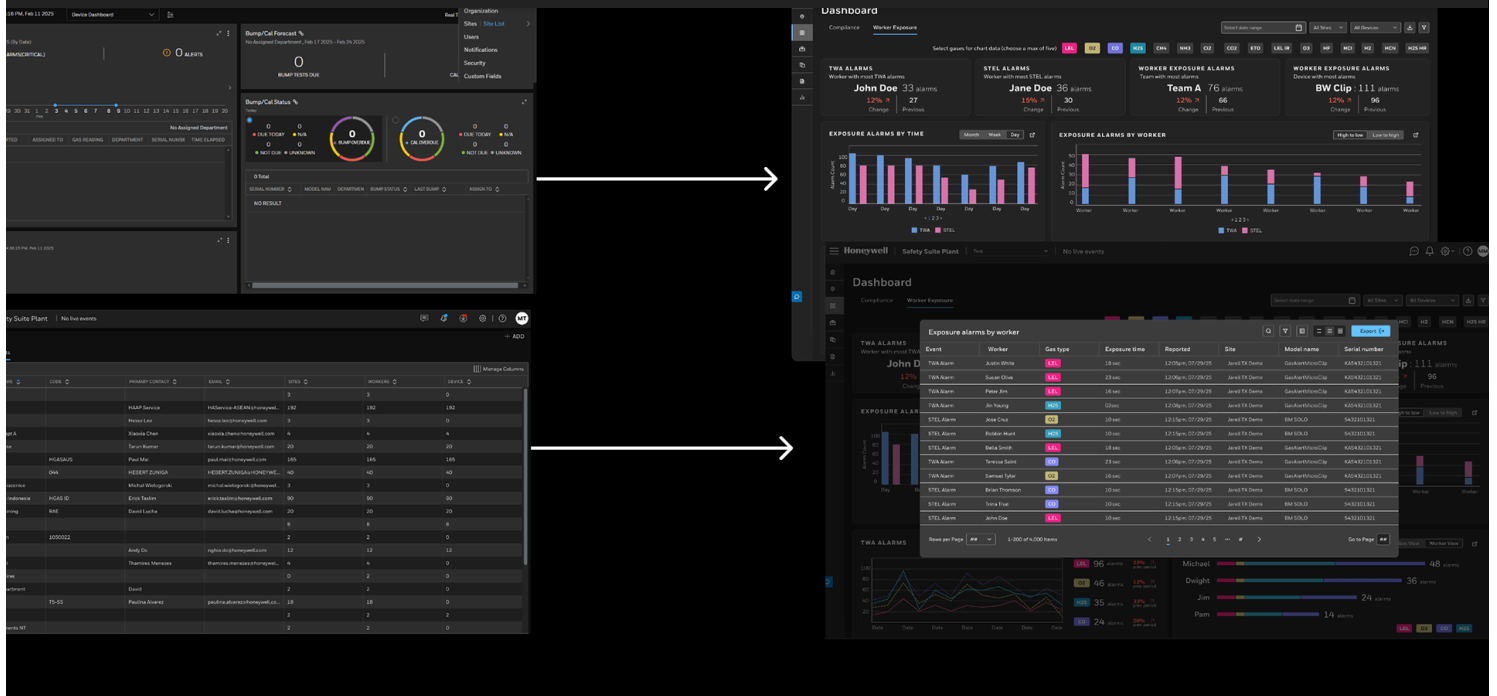
Challenge and Context

The core organizational challenge was the impending failure of Honeywell's legacy software, Safety Suite, which served as the fleet management backbone for portable and fixed gas detectors across high-risk industries (e.g., Oil & Gas, Chemicals). This software suffered from a critical usability deficit, posing a significant risk to customer safety and market share:

- **Prohibitive Setup Times:** Connecting devices like the MultiRAE to Safety Suite often took over three hours across multiple days due to complex instructions, undocumented issues (like needing an antiquated RS232 adapter cable), and over 30 confusing steps.
- **Connectivity and Data Integrity:** Customers experienced frequent failures in Bluetooth (BLE) pairing with mobile apps (Safety Communicator), data inconsistency due to time synchronization errors between devices and docks, and slow data downloading (30-60 minutes for historical logs).
- **Lack of Proactive Insight:** The existing software lacked true analytics, preventing customers from gaining visibility into asset status, user accountability (who missed bump tests/calibrations), and the root cause of incidents.

This situation demanded a complete digital transformation to create an enterprise safety ecosystem (EWAS) that prioritized user experience and technological reliability.

Matt McGlothlin



Design Goal and Assumptions

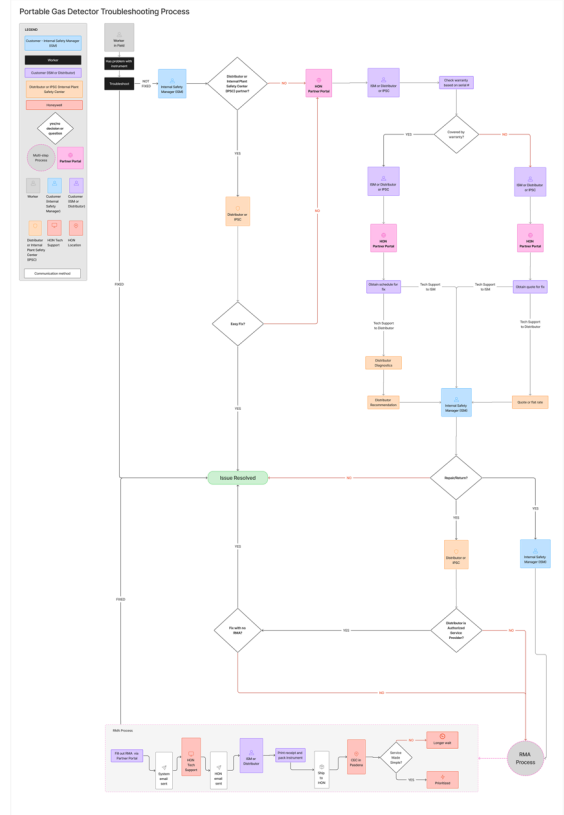
Transform a fragmented legacy ecosystem into an intelligent, connected experience that empowers industrial workers, safety managers, and technicians

I led the user experience strategy and design while defining the onboarding flow, AI interactions, and dashboard systems, and helping shape the integration of Honeywell Forge 2.0 across the new experience.

As Lead UX Designer I was able to spearhead transformation by designing a next-generation enterprise gas detection ecosystem (EWAS) that delivered seamless device setup, modernized fleet management, assured regulatory compliance, and laid the foundation for AI-enabled safety services.

Solve the Usability Crisis: Create intuitive and dependable mechanisms for device onboarding and synchronization (replacing the previous 30+ step failure point).

Enable Predictive Operational Intelligence: Integrate AI/ML insights into a Command-and-Control style interface to forecast compliance issues, predict leaks (gas plume modeling), and proactively suggest safer worker routes.



outcomes, specifically addressing lone worker scenarios and emergency response readiness.

Definition - Understanding the Problem

In my role as lead designer, I collaborated with the UX research lead to systematically apply human-centered design techniques including on-site workshops, stakeholder design collaborations, observational voice of customer, competitive analysis, usability testing and prototyping.

Diagnostic Research and Constraint Mapping

We conducted extensive UX domain research, mapping the existing dysfunctional workflows and performing an internal HUE information architecture test that meticulously documented the pain points of the legacy Safety Suite installation/setup process. Simultaneously, we benchmarked against top competitors (Blackline, Dräger, MSA) to define "table stakes" features like device assignment, compliance data and real-time monitoring, informing the competitive strategy for EWAS.

Current state workflows

I deconstructed Safety Suite's existing screens and workflows.

- Device setup and configuration
- Check-in / Check-out (device assignment)
- Fleet management
- Real-time monitoring
- Safety outcomes and reporting

This analysis revealed deeply nested flows and redundant paths as many users had to switch between desktop apps, mobile apps (Safety Communicator, Device Configurator), and docking stations just to perform simple tasks like firmware updates or alarm calibration.

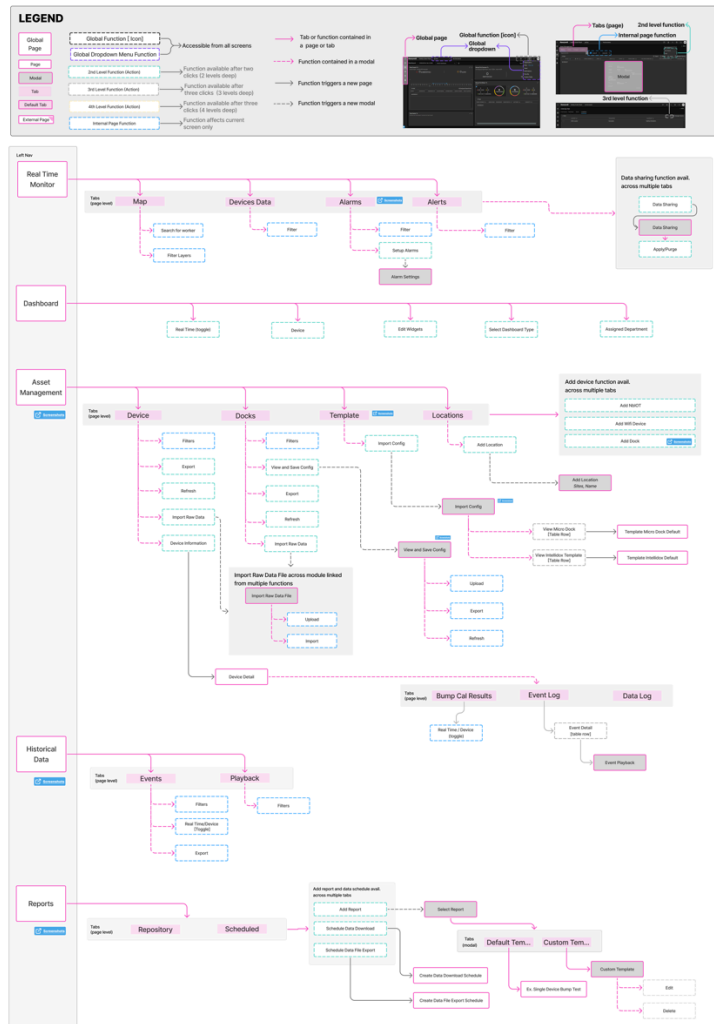
Key friction points identified:

- Inconsistent BLE pairing experiences

Onboarding guidance

- Fragmented data between fleet health, device usage, and safety outcomes

Safety Suite Information Architecture (current state)



Mapping the Legacy System

Before jumping into design, I needed to fully understand how Safety Suite’s legacy experience functioned — not from documentation, but from the user’s lived perspective.

To do that, I mapped the entire Safety Suite ecosystem using a unified visual language of boxes, arrows, and a custom legend.

Each color and symbol represented a specific interaction type:

- Pages – primary views or dashboards
- Modals – secondary, task-based interactions
- Navigation paths – directional arrows showing user journeys
- Tabs – nested content areas
- Subprocesses – loops or dead-end flows where users often got stuck

This model became a “single source of truth” for understanding how all modules connected — including Device Configurator, Fleet Manager, Real-Time Monitoring, and Safety Outcome.

Findings from the Workflow Map

- The system had overlapping entry points for similar tasks (e.g., device configuration accessible from multiple menus)
- Key workflows like “Assign Device” and “Add Site” were buried two or three layers deep, often behind modal dialogs
- The navigation hierarchy wasn’t role-based as managers, technicians, and admins viewed the same menus, creating cognitive overload
- Multiple redundant loops added unnecessary clicks and confusion

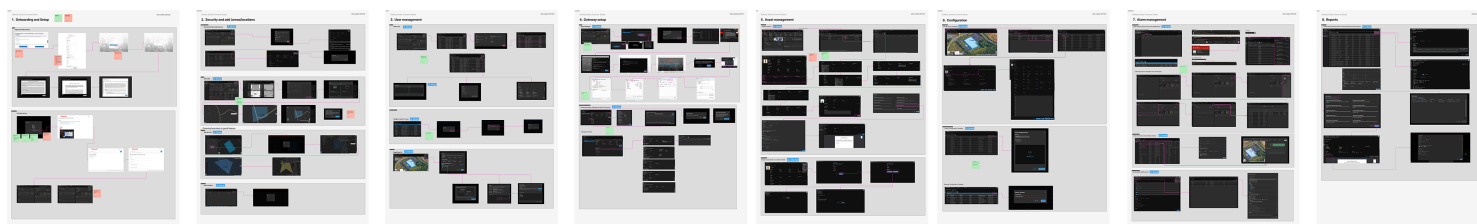
This mapping exercise clarified why users felt lost. It wasn’t because the product lacked capability, but its architecture had grown

I created a clickable system map as a benchmark for simplification and as a collaboration artifact.

Benefits of the map included:

- Engineering alignment on what could be consolidated under Forge 2.0
- Demonstrations during workshops to show where friction occurred
- Annotated with insights (from VOC and app store reviews) to pinpoint pain points and prioritize redesign areas

This diagram directly informed how we structured the new EWAS navigation model by collapsing redundant pages into intuitive modules and introducing a guided setup path for first-time users.



Research - Competitive Analysis and Benchmarking Best-in-Class Safety Systems

The UX team and I compared Honeywell's experience with leading safety platforms such as MSA Grid, Blackline Live, Dräger Gas Detection Connect, Industrial Scientific iNet, and WatchGas SST.

Each had distinct advantages:

- MSA: RFID/NFC-based digital check-in/out
- Blackline: Real-time cloud dashboards and analytics
- Dräger: Cloud-based monitoring with role-based permissions
- WatchGas: Compliance kiosks with NFC tap-to-assign simplicity

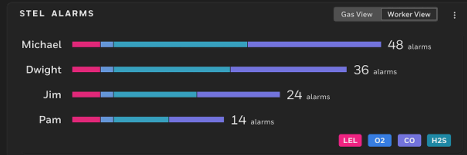
By plotting these differentiators against Honeywell's current UX, we identified four experience gaps to address:

1. Simplified onboarding (reduce steps and technical jargon) that would be worker-centric
2. Smart device assignment (NFC/QR tap-based pairing)
3. Unified visibility (fleet + worker + compliance in one dashboard) with accessibility at the forefront
4. Guided steps through complex workflows supplemented with AI assistance (reduce dependency on support)

a) Worker-centric

○ ASSIGN TO USER

○ ASSIGN TO TEAM



SCUF / HCUI

- Last update in 2022.
- Rendering issues due to different React versions.
- Not accessible.



b) Easy, fast data capture

Max file size should not exceed 10MB*
Supported files are (insert file type)

Serial Numbers CSV

Serial Numbers Added

5220BWF05232000056	✓
5220BWF05232000057	✓
5220BWF05232000058	✓

Scan the QR code
The QR code will automatically
be detected when you position
it between the guidelines.

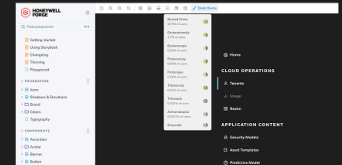


Ready to scan
Place the device for reading
behind your phone.

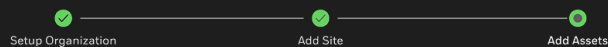


Forge 2.0

- Supported by Honeywell Connected Enterprise.
- Accessibility compliant.
- Missing mobile components.



c) Guiding complex workflows



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Discovery - Validating Pain Points with Usability Testing

With the gaps initially identified through desktop research and competitive analysis, we conducted usability walkthroughs with field technicians, product SMEs, and support engineers using the current Safety Suite. This helped us validate initial assumptions about setup.

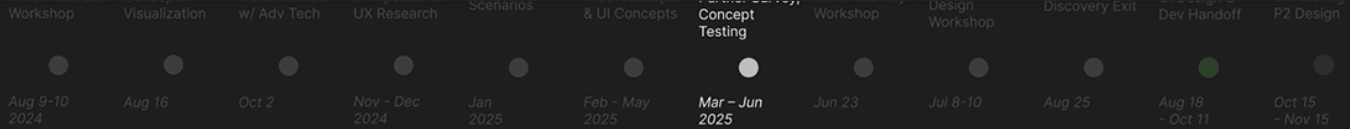
Sessions included real-world tasks such as:

- Connecting a BW Flex or MicroRAE via BLE
- Assigning a worker to a device
- Exporting calibration data

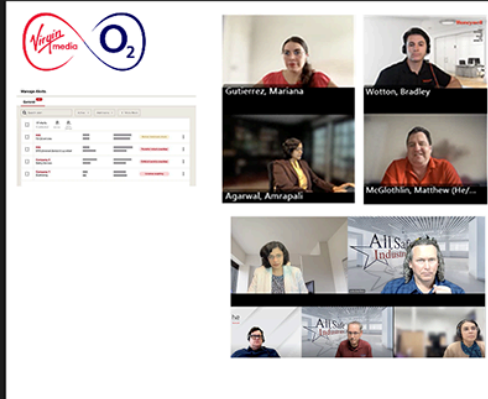
Field workers validated initial assumptions about onboarding and we uncovered new usability gaps based on UX principles and heuristics:

- Error handling was weak as users often abandoned the setup when pairing failed
- System feedback was unclear (“Success” messages appeared even when the connection wasn’t established)
- Navigation between modules (Config, Fleet, Outcomes) lacked hierarchy or a clear mental model
- Users expressed frustration that “I never know if I did it right”

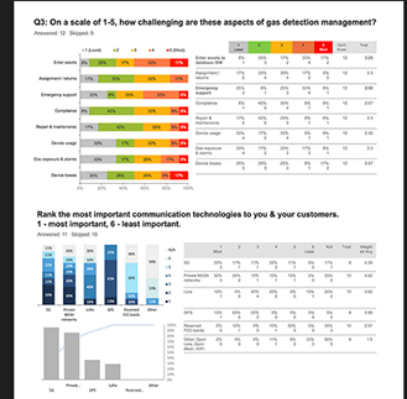
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OVOC with NOBL KD Analytical (Gov Distributor - Lexington KY)



Remote VOC with Virgin Media (Telco customer, UK) and Ailsafe (USA Partner)



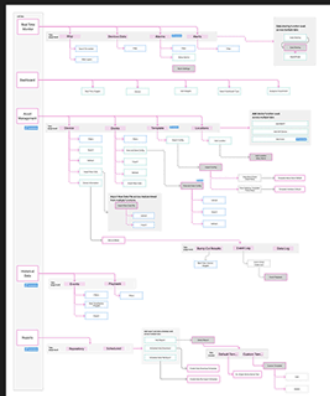
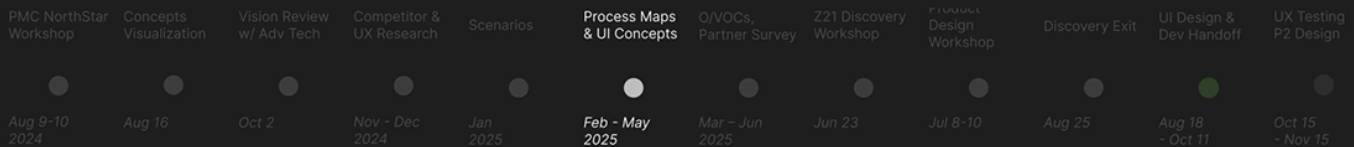
Survey with key Channel Partners

Honeywell

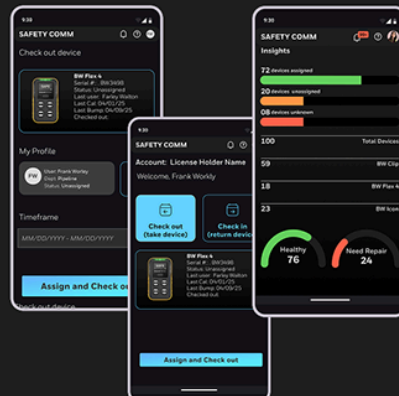
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Enterprise Worker & Asset Safety

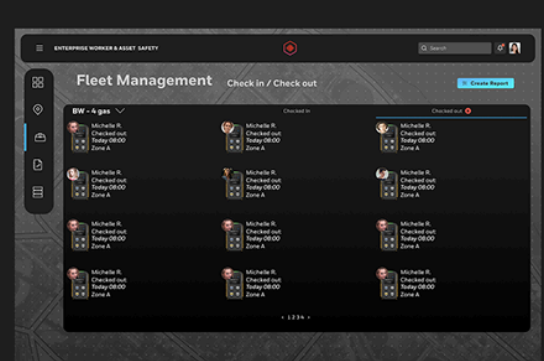
USER EXPERIENCE MILESTONES



Mapping the existing Information Architecture (Safety Suite)



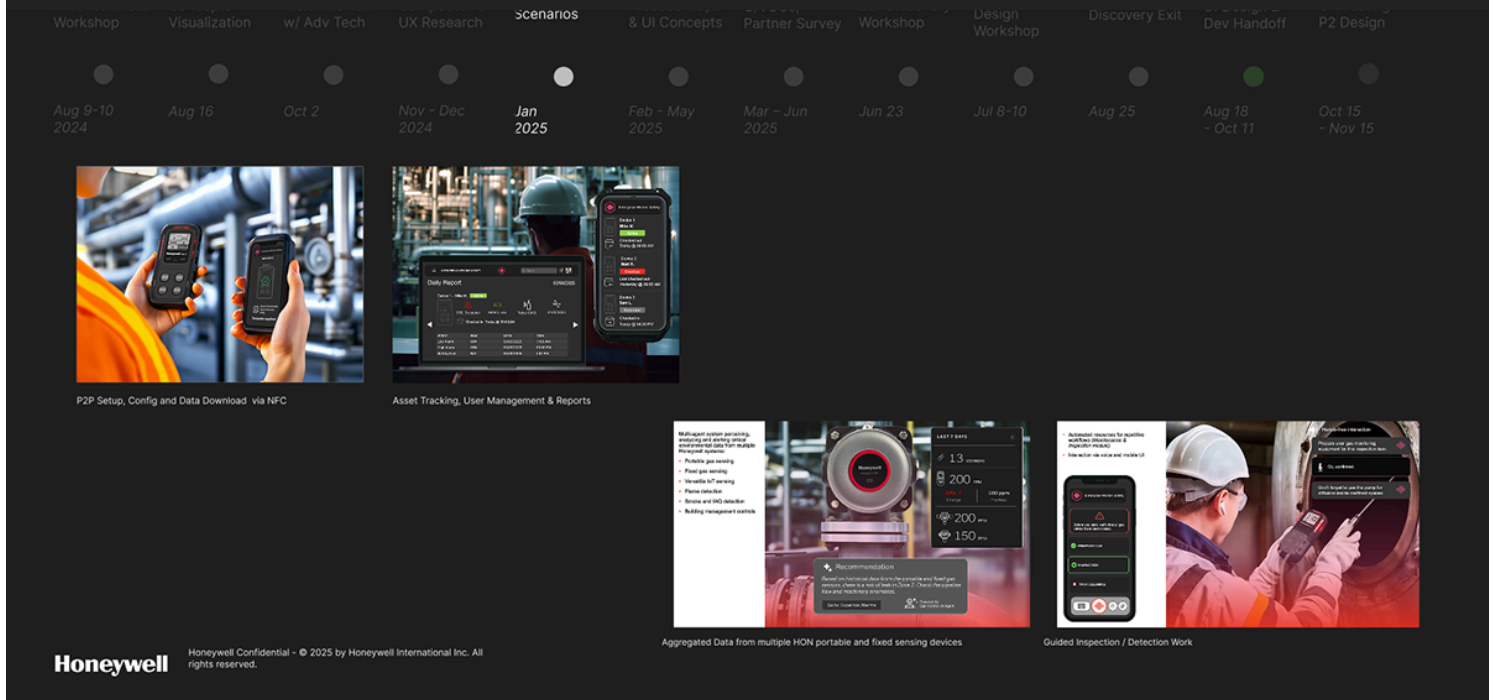
Mobile app feature: Self-assigning/unassigning assets



Desktop feature for Check in/ Check out dashboard and Report generations

Honeywell

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Designing the Onboarding Experience

We completely reimagined how devices and users join the platform.

The old multi-step manual setup became a guided, adaptive onboarding journey informed by pre-qualification questions and real-time connectivity detection.

Highlights

- **Pre-qualification flow:** Users answer short setup questions, such as site type, worker roles, real-time needs, to generate a tailored setup sequence.
- **Smart connection detection:** Automatically recommends the best connection type (gateway, mobile, or wired).
- **AI-guided troubleshooting:** Provides instant feedback when a connection fails — suggesting likely fixes and tutorials.
- **Interactive onboarding tour:** Contextual help and guided walkthroughs reduce training time for new users

Home

Real Time Map

Dashboards

Asset Management

Fleet Management

Reports

Historical Data

Welcome Richard, let's get setup ...

Exit Onboarding

Setup Organization Add Site Add Assets

Setup Initial Site

This will be your default site. It can be same as your organization location details.

All fields required

Site Name

[Company name duplicated from sign up]

Address

[Address duplicated from sign up]

No P.O. Boxes

City

[Duplicated from sign up]

State/Province

[Duplicated from sign up]

Zip/Postal Code

[Duplicated from sign up]

Country

USA

+ Add New User

Previous

Next

GUIDED ONBOARDING

Honeywell

Safety Suite

Help Notifications Settings Profile MM

Home

Real Time Map

Dashboards

Asset Management

Fleet Management

Reports

Historical Data

Welcome Richard, let's get setup ...

Exit Onboarding

Setup Organization Add Site Add Assets

Setup Organization

Enter organization details

All fields required unless marked as optional

Company Name

[Company name duplicated from sign up]

Industry

Select an option

Website - Optional

Enter website

Address

[Address duplicated from sign up]

No P.O. Boxes

Country

USA (+01)

Phone Number

[Duplicated from sign up]

City

[Duplicated from sign up]

State/Province

[Duplicated from sign up]

Zip/Postal Code

[Duplicated from sign up]

Next

Welcome to EWAS+

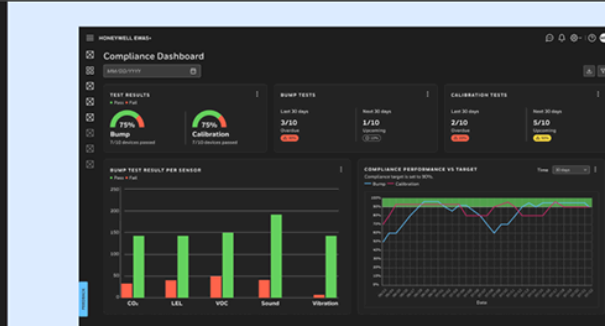
Experience EWAS+ with smarter insights, faster response, and improved usability. Enjoy enhanced monitoring and analytics during this trial - upgrade to keep full access to these advanced features.

Onboarding Wizard

Compliance Dashboard

Worker assignment

Configuration



The Compliance Dashboard gives you clearer, smarter insights into your devices. With improved analytics and intuitive visuals, it's easier than ever to track status, spot issues, and stay compliant at a glance.

Skip all

Next

ACCOUNT CREATION & SIGN IN

New Users:

Existing Users:



Create account

All Fields Required

First name Last name

Company

Country Code Phone

Address

City State/Province Zip or Postal Code

City State/Province Zip or Postal

Email

name@company.com

Password

Confirm password

☐ Remember me

Create account

Already have an account? [Log in](#)

[Privacy](#) | [Cookies](#) | [Terms](#)

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Honeywell Safety Suite

Log In

Email or username

john.doe@honeywell.com

Password

☒ Remember me

Log in

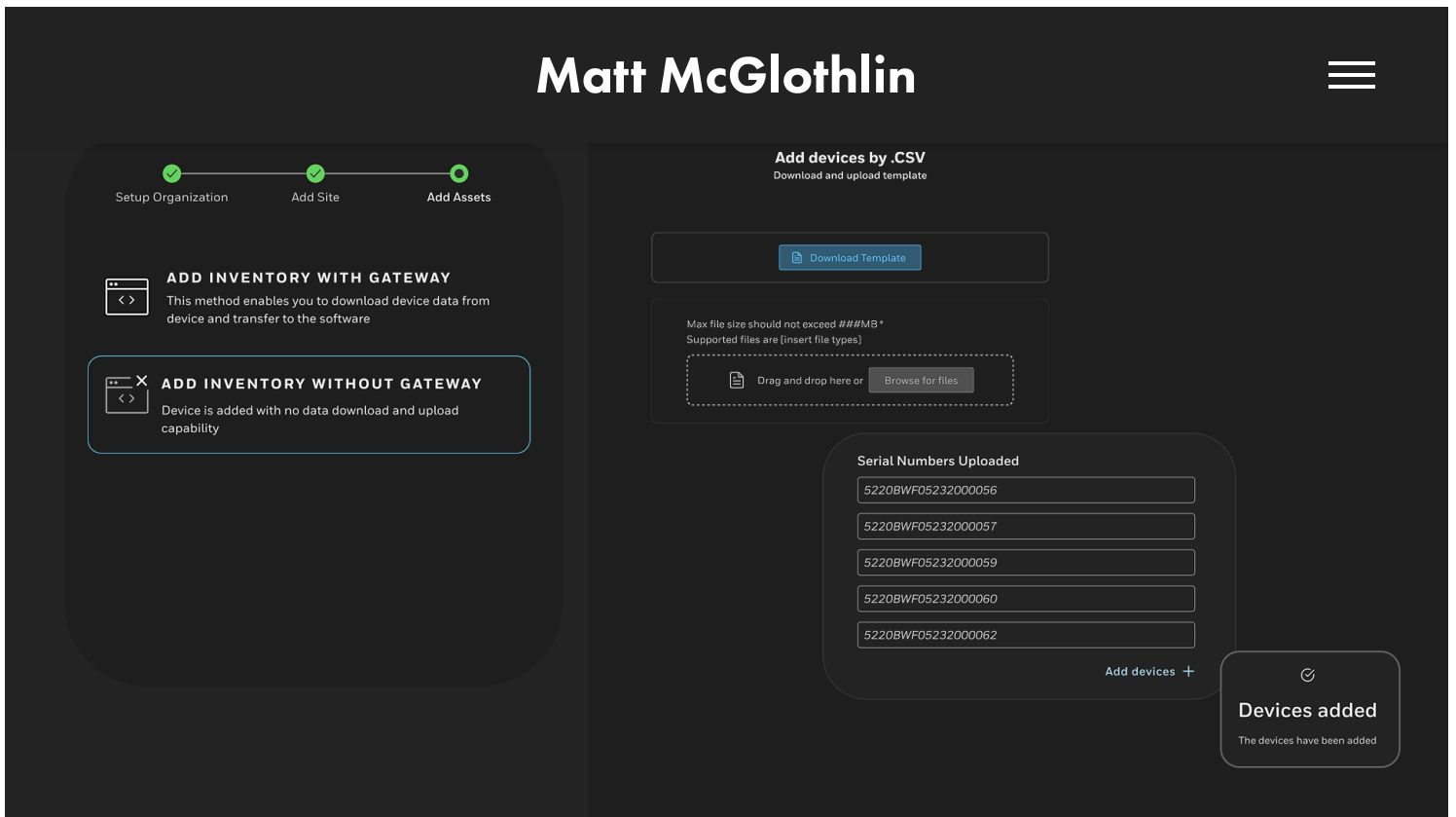
[Use single sign-on](#)

[Forgot password](#)

[Don't have an account?](#) [Create account](#)

[Privacy](#) | [Cookies](#) | [Terms](#)

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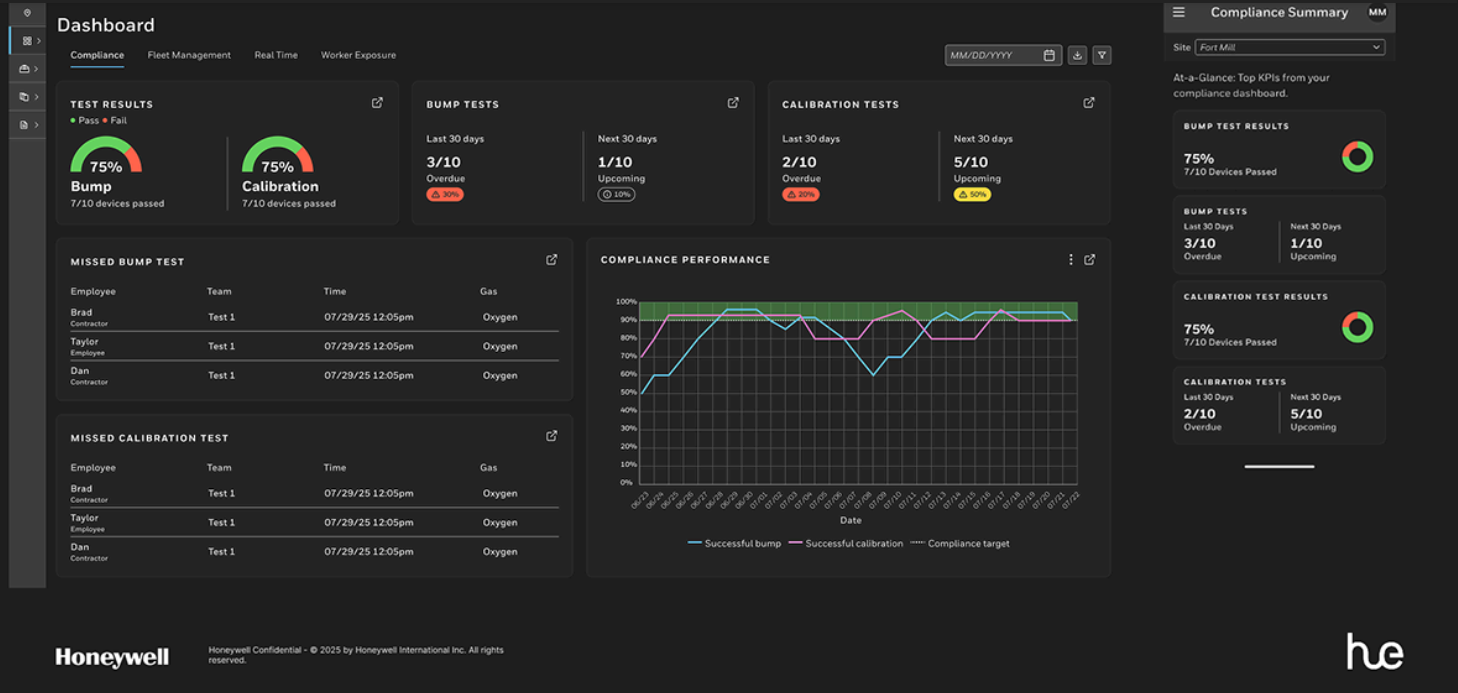
Fleet & Asset Management Redesign

We rebuilt the Fleet Management dashboard with real-time visibility displaying:

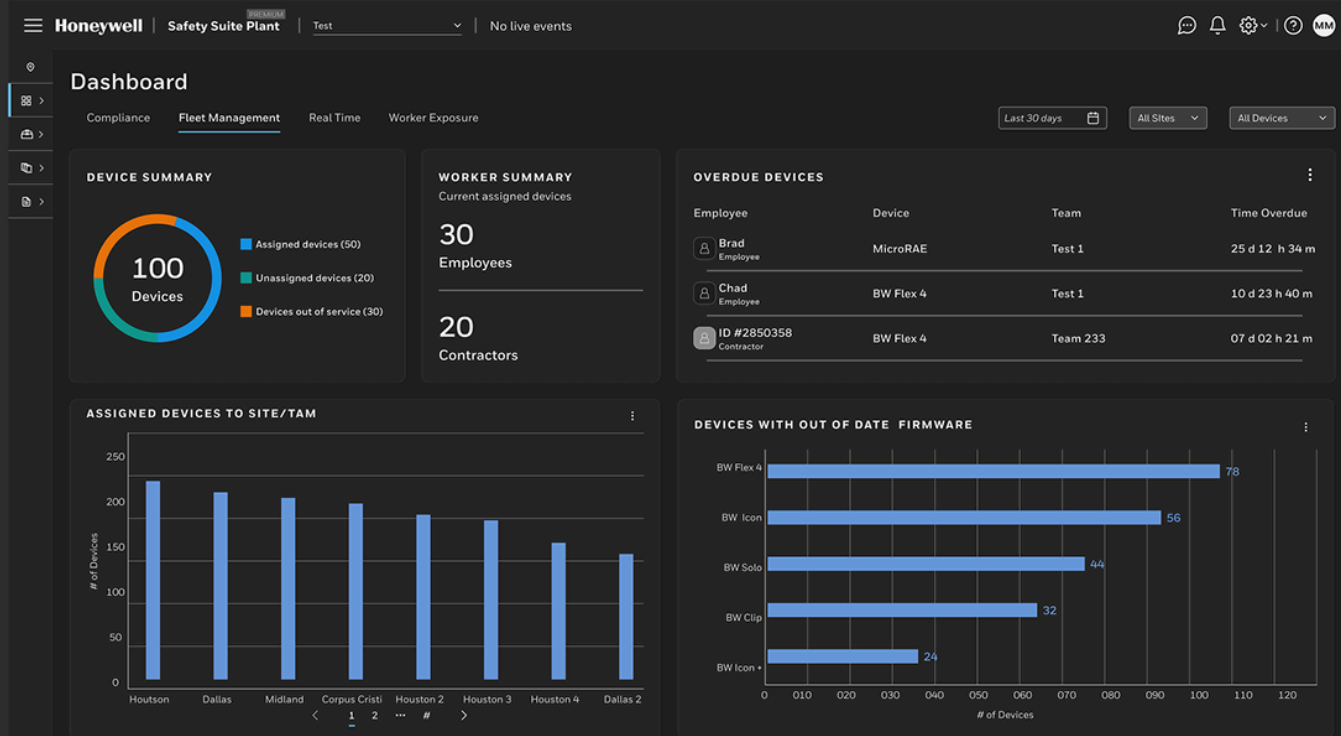
- Device status, calibration health, and battery life
- Worker assignments and site locations
- License management and compliance summaries

The dashboard leverages Honeywell Forge 2.0 components for data consistency and performance, blending cloud analytics with edge intelligence for low-latency insights.

Matt McGlothlin



FLEET ASSIGNMENT INSIGHTS



AI in the Experience

Artificial Intelligence served both as a design partner and a product capability.

MS Copilot Notebooks and Miro AI to accelerate ideation

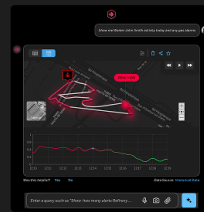
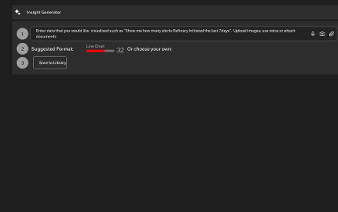
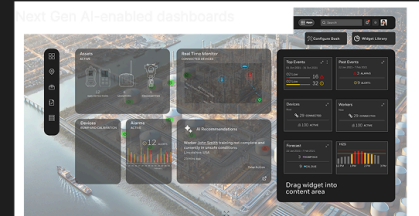
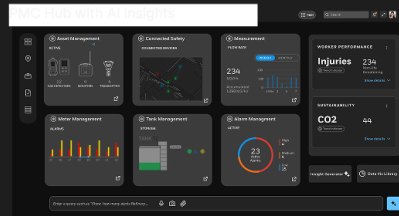
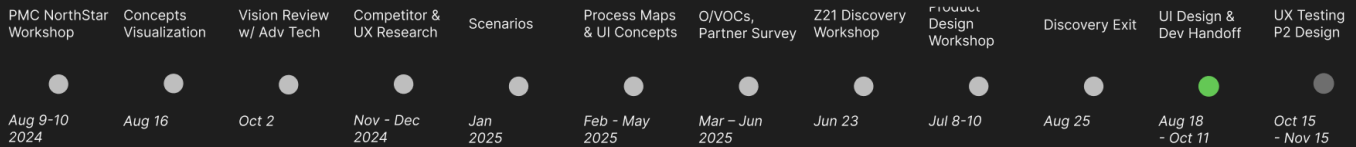
- Upload documentation and presentation to a custom notebook that served as a research assistant

Artificial Intelligence in the product

- Maintenance Assist: Context-aware AI that explains setup errors, recommends configuration sequences, and assists in troubleshooting.
- Predictive Analytics: AI models predict hazardous situations by analyzing data logs from devices and external environmental factors
- Behavioral AI: Worker training and compliance data (device bump tests and calibrations) analyzed by AI to determine and predict likelihood of injury

Enterprise Worker & Asset Safety

USER EXPERIENCE MILESTONES



Honeywell

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NFC / QR Innovation

A key competitive differentiator was introducing NFC and QR-based device assignment.

How it Works

- Workers tap or scan a detector to instantly assign it before a shift.
- Auto check-in/out updates worker-device pairing in real time.
- The system syncs with cloud fleet management even when offline.

Benefits

- Eliminated manual entry errors
- Improved accountability
- Simplified shift handoffs
- Strengthened compliance traceability

Assign to user

Select a user Select a device Select Date/Time

Select a device

Mike Mahoney
Group: Team A
Last use: 04/02/25
Status: Unassigned

Unassigned devices

Assigned devices

Serial #: 3623872388 BW Flex 4 Status: Unassigned Last user: Joe Matheny	Serial #: 3623872388 BW Flex 4 Status: Unassigned Last user: Joe Matheny	Serial #: 3623872388 BW Flex 4 Status: Unassigned Last user: Joe Matheny
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Previous

Next

Assign by device

Site Fort Mill

Review
Verify your selections, add shift details.

SELECTED DEVICE
 BW Flex 4
Serial Num: M12729304
Status: Unassigned

SELECTED TEAM
 Team CD
9 Devices Assigned

Date range - Optional
MM/DD/YYYY - MM/DD/YYYY

Hours - Optional
8 hours

Next

OVERVIEW OF DEVICES AVAILABLE & ASSIGNED

Honeywell | Safety Suite Plant | Test | No live events

- Home
- Real Time Map
- Dashboards
- Asset Management
- Fleet Management
- Reports
- Historical Data

Assignment Overview

Available Devices 32 Devices assigned to users 1 Devices assigned to teams 1

FILTERS Site Select an option ☐ Single gas detector ☐ Multi-gas detector Make/Model BW Flex

Serial #: 3623872388 BW Flex 4 Status: Unassigned Last user: Joe Matheny	Serial #: 3623872388 BW Flex 4 Status: Unassigned Last user: Joe Matheny	Serial #: 3623872388 BW Flex 4 Status: Unassigned Last user: Joe Matheny
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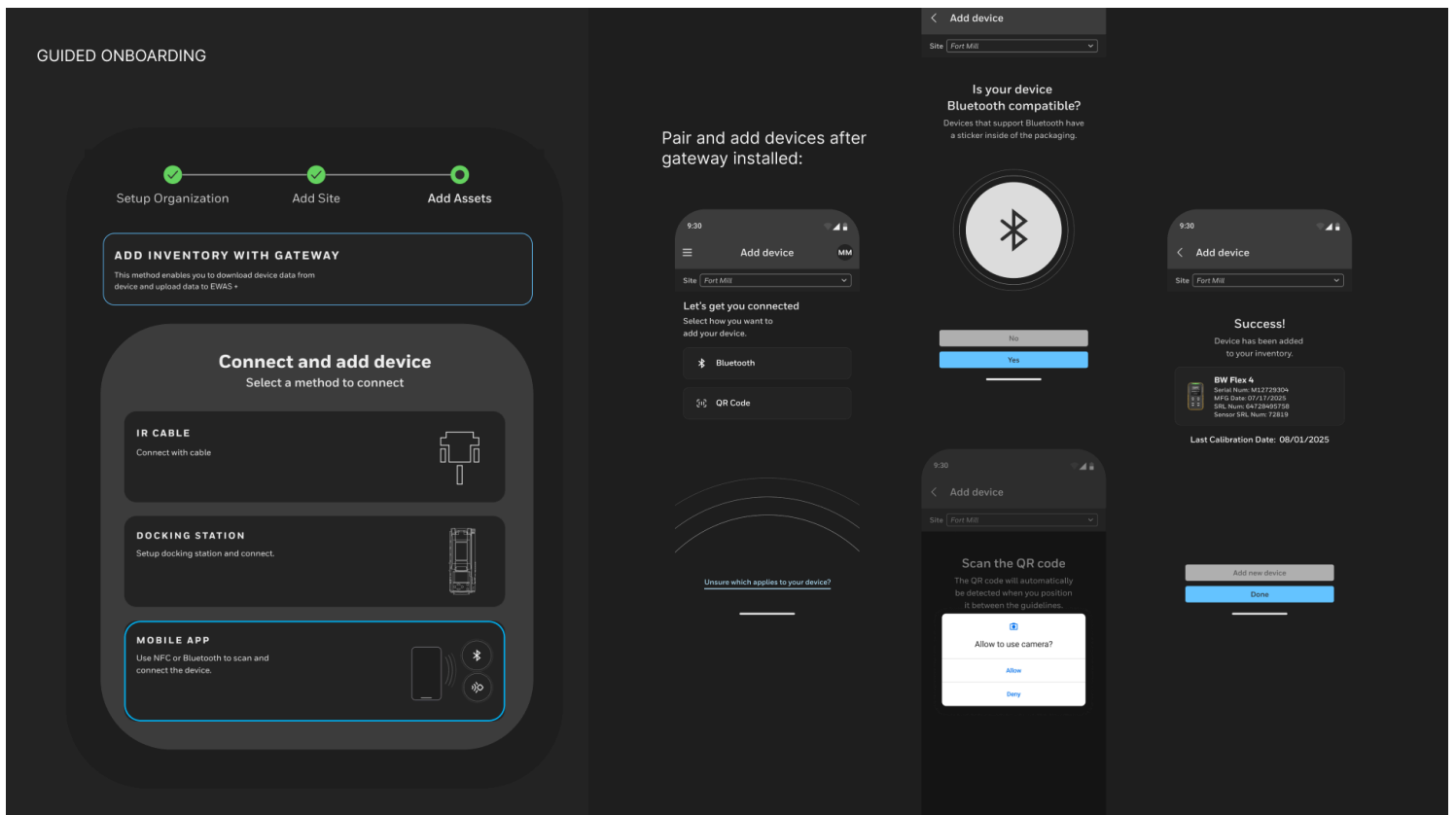
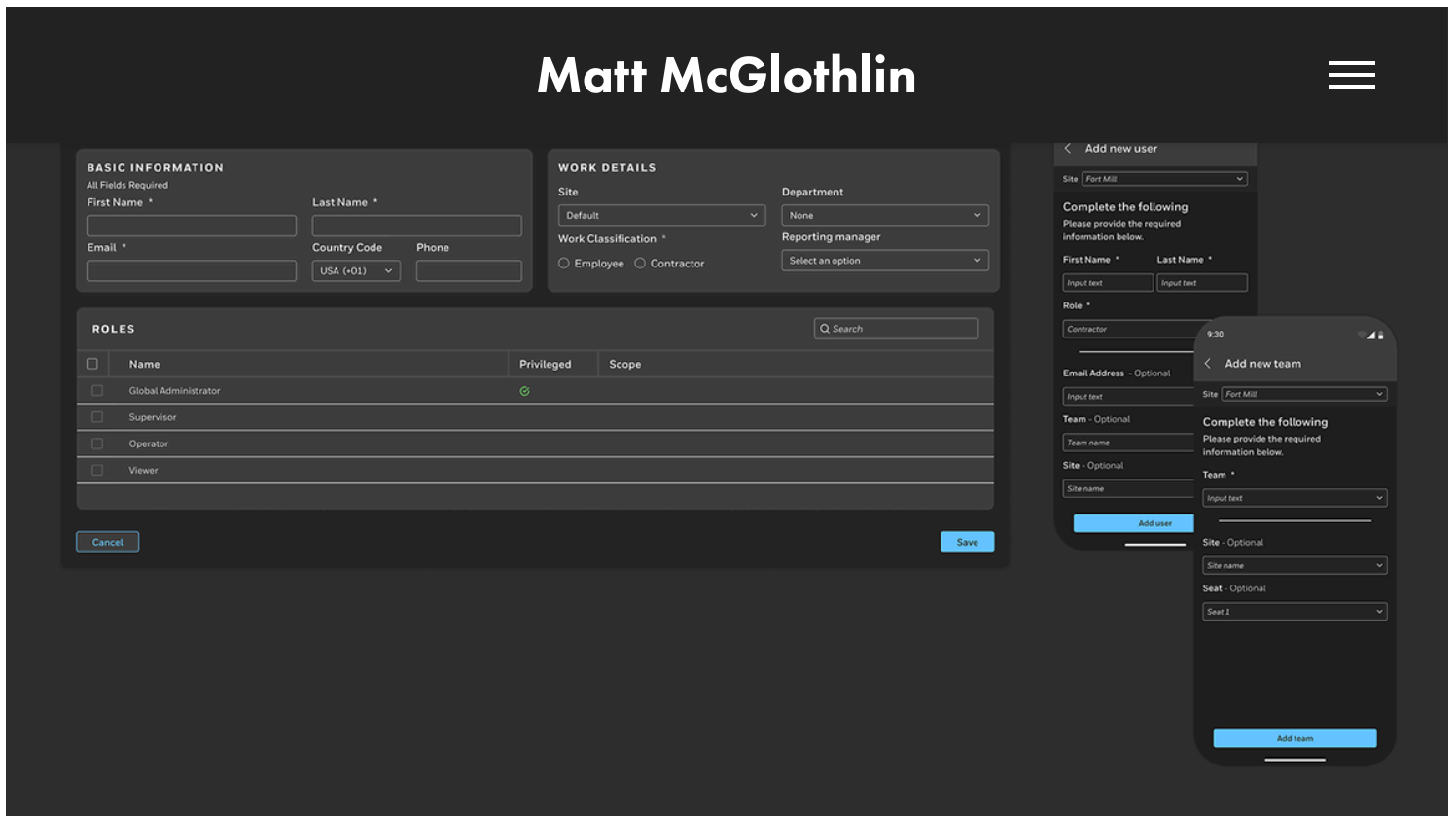
Cancel

Add new team

Add new user

Assign

FEEDBACK



Design System & Dashboards

We implemented the Honeywell Forge 2.0 Design System for visual and interaction consistency:

- Modular cards and analytics widgets
- Accessible color palette and high-contrast data visuals

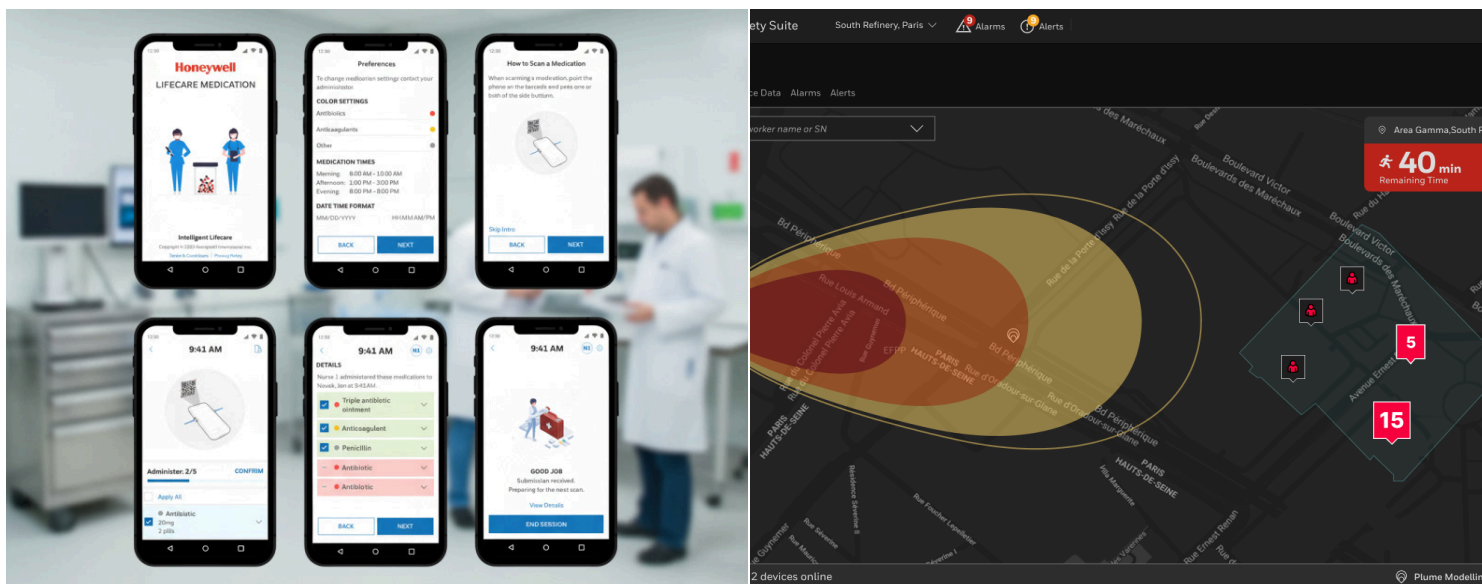
Reflection

This project taught me that AI in industrial design isn't about automation — it's about augmentation. By merging research, empathy, and intelligent systems, we created a more human, transparent, and scalable way for teams to stay safe and productive.

Next Steps

- Integrate Maintenance Assist directly into Forge
- Expand NFC/QR to fixed detection devices
- Launch AI-driven compliance reports and predictive maintenance modules

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